

Class 38: Markov Chains

Review: What is an eigenvector for a square matrix A ?

Connections

Markov Chains are mathematical models that describe the probability of future conditions (states) based on previous states and a transition matrix.

Example: (from *Introduction to Linear Applications* by DeFranza and Gagliardi)

A group insurance plan allows three different options for participants, plan A, B, and C. Suppose that the percentages of the total number of participants enrolled in each plan are 25 percent, 30 percent, and 45 percent, respectively. Also from past experience assume that participants change plans as shown in the table.

		From		
		A	B	C
T_0	A	.75	.25	.2
	B	.15	.45	.4
	C	.1	.3	.4

Transition Matrix

Graph representation

Current state vector $V = \begin{bmatrix} .25 \\ .30 \\ .45 \end{bmatrix}$

The state vector after one transition is given by multiplying Tx .

What is the state vector after five years?

Find the state vector after ten years.

Find the state vector after ten years if the percentages are 20, 25, and 45 at the beginning. Try any other probability vector as a starting state.

Try 48, 30, and 22 percent as a starting state.